

Mode 7 — Baseline-BFP

DP16 (16×8) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents

Step 1 — DP16 (16×8)

$$\text{DP16 (16} \times 8) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP16 (16} \times 8) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 (16} \times 8)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 (16} \times 8)}$$

Step 3 — DP8 (16×4)

$$\text{DP16 (16} \times 8) = 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (16} \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (16} \times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP16 (16} \times 8) &= 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hi}}_{\text{DP8 (8} \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lo}}_{\text{DP8 (8} \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hi}}_{\text{DP8 (8} \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lo}}_{\text{DP8 (8} \times 4)} \\ &\quad + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hi}}_{\text{DP8 (8} \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lo}}_{\text{DP8 (8} \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hi}}_{\text{DP8 (8} \times 4)} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lo}}_{\text{DP8 (8} \times 4)} \end{aligned}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4))

$$\left(2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{hi} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{lo} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{hi} + \sum_{i=0}^7 a_i^{lo} b_i^{lo}, E_0 \right), \left(\dots i \in [8, 15] \dots, E_1 \right) \longrightarrow (M, E)$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{DP16 (16} \times 8) = M \cdot 2^E$.

Step 6 — Result

$$\begin{aligned} M &= 2^{-\delta_0} \left(2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{hi} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{lo} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{hi} + \sum_{i=0}^7 a_i^{lo} b_i^{lo} \right) \\ &\quad + 2^{-\delta_1} \left(2^{12} \sum_{i=8}^{15} a_i^{hi} b_i^{hi} + 2^8 \sum_{i=8}^{15} a_i^{hi} b_i^{lo} + 2^4 \sum_{i=8}^{15} a_i^{lo} b_i^{hi} + \sum_{i=8}^{15} a_i^{lo} b_i^{lo} \right), \\ E &= \max(E_0, E_1) \end{aligned}$$